

Q1 - Fight Duration Over Time (UFC - fight duration)

Looking at the fight duration timeline, the introduction of the Unified Rules in 2001 clearly splits the data into two distinct eras. During the first seven years, the UFC lacked weight classes (until 1997), scorecards, rounds, and consistent time limits. Fighters often competed multiple times in a single night, and matchmakers focused on putting different martial arts styles against one another.

Bouts could end in two minutes, like Royce vs. Jimmerson, or drag on for 36 minutes, as seen in Royce vs. Ken Shamrock. Because of these shifting rules, varying fighter backgrounds, and small annual samples, both the mean and median durations swing wildly across the early graph. Royce Gracie is a prime example of this chaos, submitting multiple opponents in a single night at UFC 1 and UFC 2 despite weighing around 80 kg in an open-weight bracket.

Quartas de final		Semifinais		Final	
 Royce Gracie (Brazilian Jiu-Jitsu)	FIN				
 Art Jimmerson (Boxe)	2:18				
 Ken Shamrock (Shootfighting)	FIN				
 Patrick Smith (Taekwondo)	1:49				
 Gerard Gordeau (Savate)	TKO				
 Teila Tuli (Sumô)	0:26				
 Kevin Rosier (Kickboxing)	TKO				
 Zane Frazier (Kenpō)	4:20				

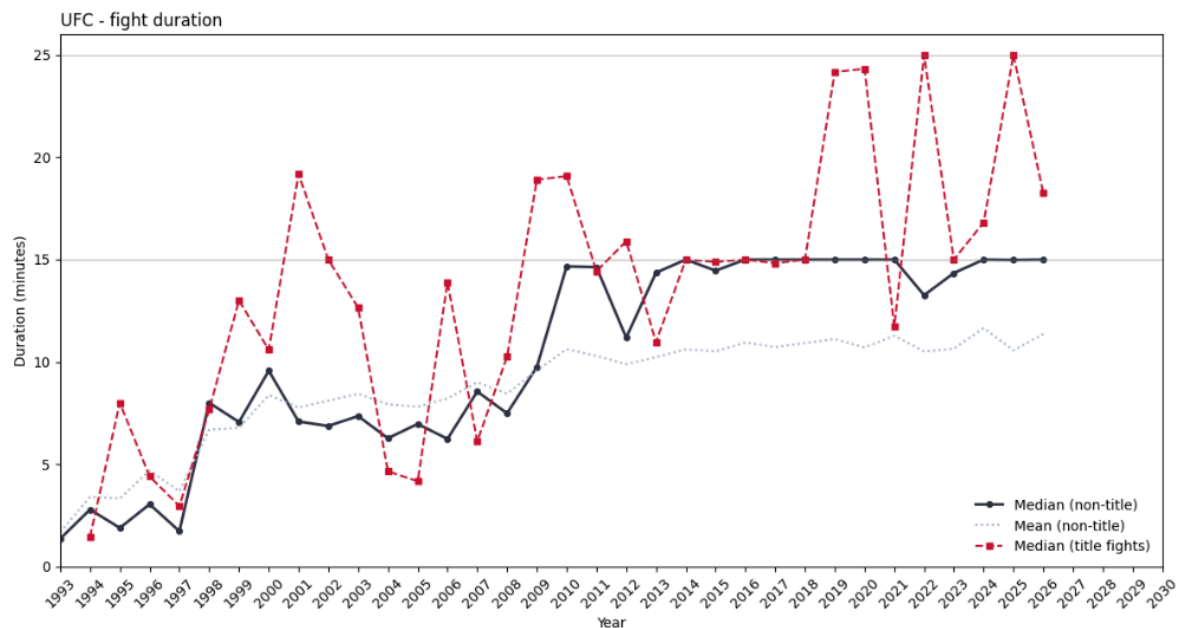
 Royce Gracie (Brazilian Jiu-Jitsu)	FIN
 Ken Shamrock (Shootfighting)	0:57
 Gerard Gordeau (Savate)	TKO
 Kevin Rosier (Kickboxing)	0:59

 Royce Gracie (Brazilian Jiu-Jitsu)	FIN
 Gerard Gordeau (Savate)	1:44

UFC 1

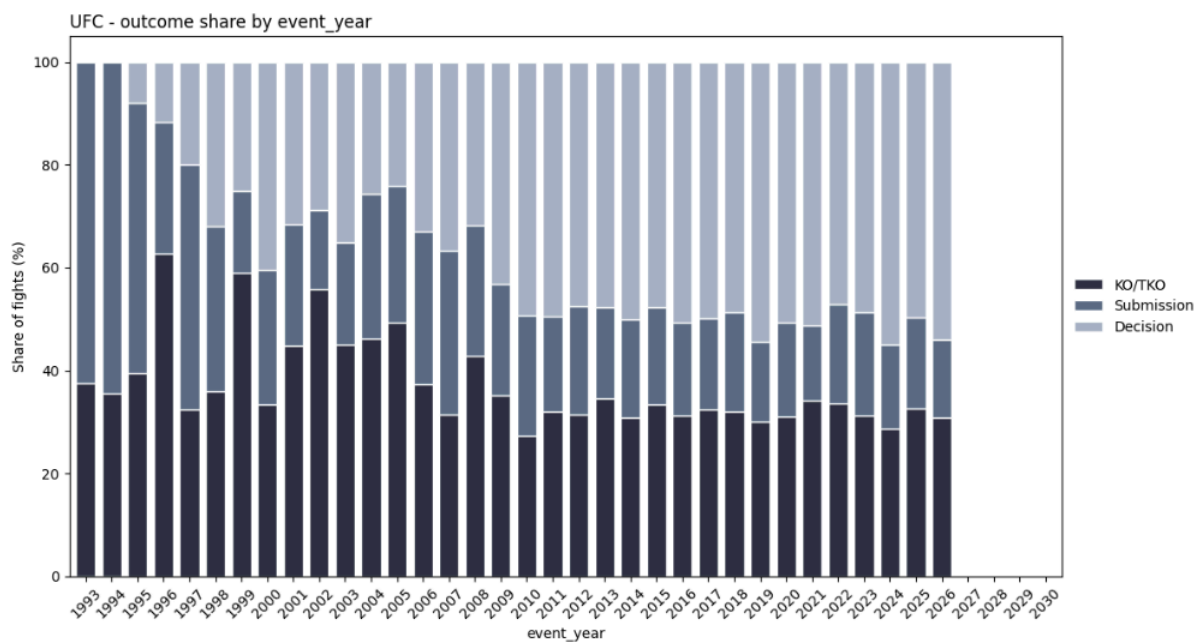
The late 90s saw transitional volatility as the UFC tried to appease regulators. Then, Zuffa bought the promotion in 2001 and adopted the Unified Rules. From that point on, the median duration climbs steadily through the 2000s and flattens out at exactly 15 minutes over the last decade.

Since regular fights max out at 15 minutes (three five-minute rounds), this median plateau is a direct result of more than half of all bouts now going the distance. The mean stays around 10 to 11 minutes because quick finishes pull the average down. For five-round title fights, the median also frequently hits its 25-minute cap. While title bout data is jumpier due to the small sample size, fighting to the final bell is now the most common outcome across all card levels.



Q2 - Outcome Distribution Over Time (UFC - outcome share by event_year)

Match results directly drive the duration trends we see above. In 1993, with no judges or real time limits, there were zero decisions. Submissions peaked heavily in 1994. Back then, pure strikers lacked grappling defense, making submissions an easy path to victory for ground specialists.

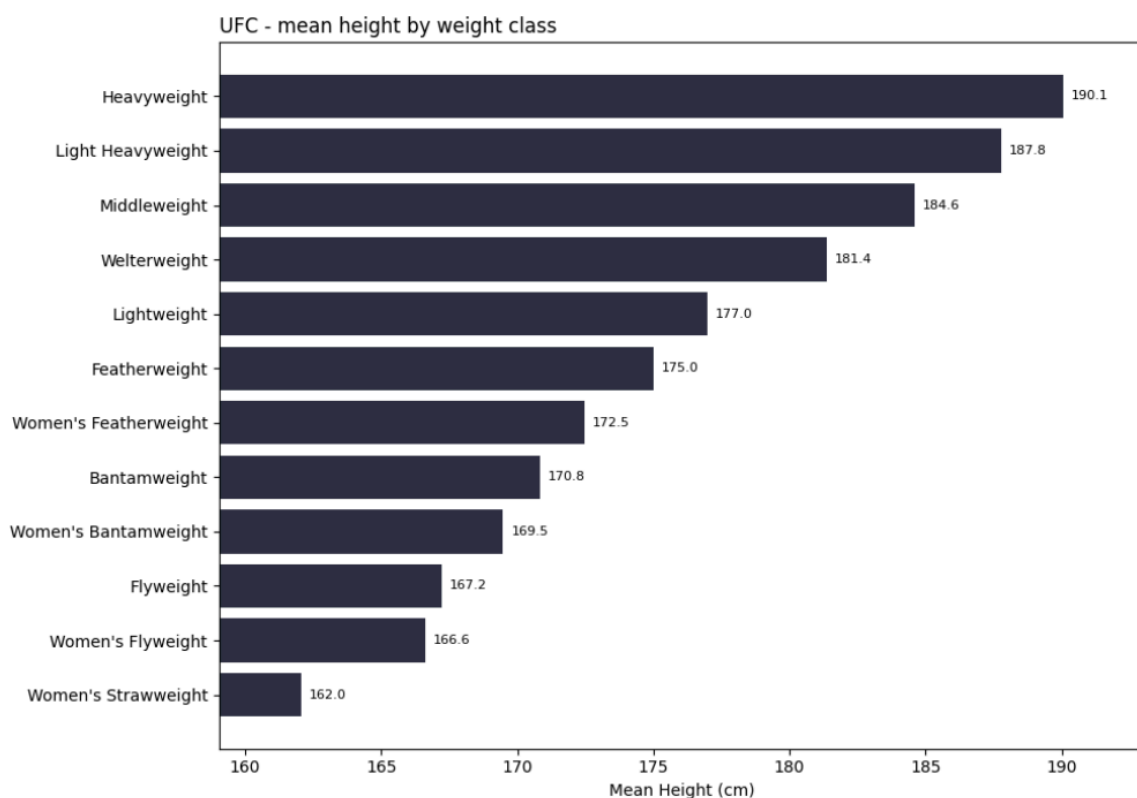


The 2001 Unified Rules changed the landscape. Time limits brought in scorecards, but athletes also evolved. As fighters rounded out their defensive abilities across the board, the submission rate steadily dropped throughout the 2000s. By 2010, the numbers settled: decisions happen in about 50% of bouts, knockouts sit between 30% and 35%, and submissions have shrunk to 15% to 20%. Modern fighters are simply harder to finish, which locks in the 15-minute decision as the standard UFC bout.

Q3 - Mean Height by Weight Class (UFC - mean height by weight class)

Average fighter height scales reliably alongside weight class limits, climbing about 3 to 4 centimeters per division from 167.2 cm at Flyweight to 190.1 cm at Heavyweight. Fighters severely dehydrate themselves for weigh-ins to compete with the largest possible frame and reach for their given category. The Heavyweight data requires a closer look. While the historical average is 190.1 cm, the division is getting taller. It is crucial to highlight that this 195 cm average specifically reflects the current UFC top 5, showing a clear physical escalation among the division's elite.

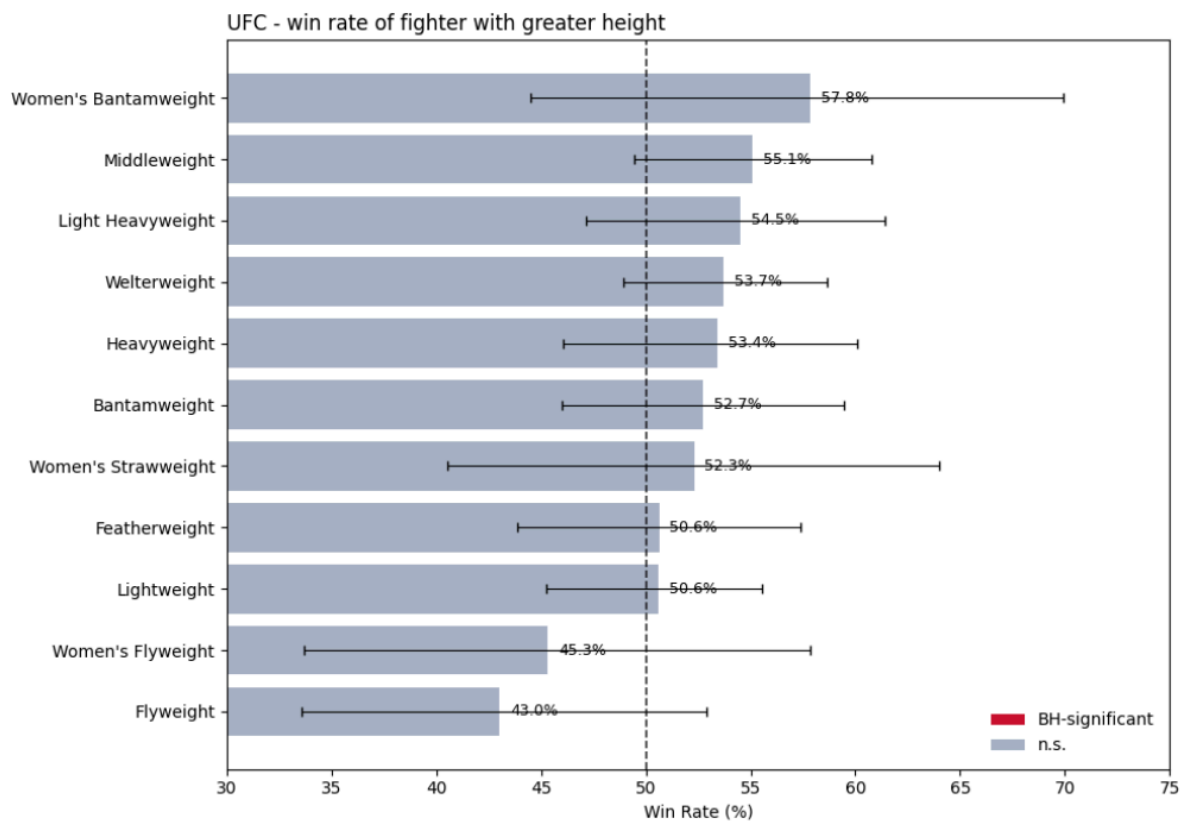
Women's divisions show a similar progression, moving from 161.6 cm at Strawweight to 172.0 cm at Featherweight. Establishing these baselines helps us evaluate whether being taller than average actually helps a fighter win.



Q4 - Win Rate by Height Advantage (UFC - win rate of fighter with greater height)

Many assume being taller provides an edge, but the data does not back that up. Looking at the 50% win rate line (essentially a coin flip), some point estimates lean toward the taller fighter (like Women's Bantamweight at 58.1%) and others favor the shorter one (Flyweight at 42.8%).

However, the horizontal error bars for every single weight class cross that 50% mark. After applying the Benjamini-Hochberg correction, none of the height advantages show statistical significance. Being the taller fighter simply doesn't reliably boost the odds of winning. Shorter fighters bring counter-advantages to the cage, like a lower center of gravity for takedown defense or better inside boxing mechanics.



Q5 - Win Rate by Reach Advantage (UFC - win rate of fighter with greater reach)

Reach tells a very similar story. In most weight classes, the win rate for the longer-armed fighter sits right around 50%, with confidence intervals crossing the neutral line just like the height metrics. A longer reach helps manage distance, but shorter frames often generate faster inside strikes and tighter grappling leverage. In some cases, like Women's Strawweight (41.8%), the shorter reach even holds a slight edge.

